

Khaja Mujahiddin

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Data Analyst

SUMMARY

Analytics Engineer with over 5 years of experience delivering financial data platforms and AI driven risk analytics. Engineered an AI powered risk intelligence solution that lifted credit approval accuracy 14%, fraud detection 19%, and cut API latency 25%. Built Power BI dashboards that accelerated reporting 16%, using Python and SQL to provide scalable, production grade insights.

SKILLS

AI/ML & Generative AI: Agentic AI Workflows, RAG Architecture, AWS Bedrock, OpenSearch Vector Search, GPT 4, Claude, LLaMA, BERT, Stable Diffusion, Prompt Engineering, AWS SageMaker, AWS Comprehend, AWS Textract, MLC LLM, GGUF, Phi 3.5 mini, Vector Indexing, pgvector, Semantic Search

Backend Engineering: Python, Java, Spring Boot, FastAPI, J2EE, Microservices Architecture, RESTful APIs, gRPC, Event Driven Architecture, Kafka, API Gateway,DDD, CQRS, AWS Step Functions, Async Processing, Circuit Breakers, Hibernate Caching

Frontend Development: React, Redux, TypeScript, JavaScript, Next.js, Vue.js, Component Based Architecture, Responsive UI Design, Dynamic Forms, API Integration, UX Optimization, Cross Browser Compatibility

Data & Databases: PostgreSQL, Aurora PostgreSQL, MongoDB, MySQL, Redis, SQL Optimization, Query Tuning, Indexing, Schema Design, Data Modeling, Replication, Sharding, ETL Pipelines

Cloud & DevOps: AWS (EC2, ECS Fargate, Lambda, S3, RDS, Aurora, CloudWatch, SQS), Azure (Functions, Service Bus, AKS, Monitor), GCP, Docker, Kubernetes (EKS), Jenkins, GitHub Actions, ArgoCD, Terraform, CI/CD Pipelines, Blue Green & Canary Deployments, Infrastructure as Code (IaC), Release Orchestration

Testing, Security & Observability: PyTest, JUnit, Mockito, Selenium, Jest, Mocha, React Testing Library, Postman, TDD, BDD, SonarQube, Pact, OWASP, OAuth2, JWT, RBAC, IAM, KMS, Secrets Manager, Prometheus, Grafana, ELK Stack, OpenTelemetry, Splunk, Azure Monitor, Log4j2, SLF4J, Agile, Scrum, Jira, Confluence

Data Analysis & Visualization: SQL, Power BI, Tableau, Excel, R, Pandas, NumPy, Matplotlib, AWS, Azure, Google Cloud, ETL, Data warehousing

WORK EXPERIENCE

Optum
AI Software Engineer

Remote, USA
Jul 2025 - Present

- Architected AI driven BNPL risk service in Python, using Pandas for preprocessing, NumPy for feature engineering, and Matplotlib for diagnostics, raising approval accuracy 14% on AWS/Azure.
- Developed secure RESTful APIs for merchant financing and customer risk profiling using Python, PostgreSQL, and Redis implementing authentication, caching, pagination, aggregation, and optimized query handling, reducing response latency by 11% across core services.
- Built scalable microservices and database driven fraud detection pipelines integrating AWS Bedrock models, PostgreSQL, Redis, and agentic AI task orchestration for anomaly detection, automated investigations, and escalation workflows, improving fraud detection accuracy, decision consistency, and operational efficiency by 19% in production financial systems.
- Delivered interactive financial analytics suite in Power BI and Tableau, supplementing Excel data extracts, to visualize repayment trends and delinquency metrics, accelerating stakeholder decisions by 16%.

Staples
Software Engineer

Hyderabad, India
Jan 2022 - Jul 2023

- Improved payment authorization success rate from 97% to 99.7% across 500+ global dealership endpoints by re architecting core financial transaction microservices using Spring Boot and implementing fault tolerant circuit breaker patterns for resilience improvements delivered.
- Reduced p95 API latency by 25% (800ms → 600ms) and accelerated financial dashboard load times for 2,000+ regional managers by optimizing PostgreSQL query execution plans and implementing Redis backed caching across distributed systems architecture.
- Engineered robust ETL pipeline using Pandas to synchronize MongoDB with PostgreSQL data warehouse, achieving 100% consistency across 40+ markets and supporting multi tenant financial reporting.
- Reduced release regression defects by 15% and compressed deployment cycles from bi weekly multi hour rollouts to continuous automated releases by overhauling CI/CD pipeline with Jenkins and Docker, improving delivery efficiency across teams significantly overall.
- Built executive dashboards in Power BI and Tableau aggregating payment performance data, enabling finance leaders to conduct trend analysis with Excel exports and drive strategic decisions.

SRIK Consulting and Services
Software Engineer

Hyderabad, India
Aug 2019 - Dec 2021

- Developed high throughput telemetry ETL workflow with Python, Pandas, and Azure Blob storage, ensuring 98.9% data propagation reliability for 50,000+ vehicle streams.
- Redesigned PostgreSQL schema partitions to form a scalable data warehouse, enabling Tableau visualizations of vehicle configuration metrics and reducing integration errors 20%.
- Centralized backend observability across distributed microservices using ELK stack and Prometheus based alerting, reducing mean incident troubleshooting time by 25% and improving system visibility and operational efficiency across production environments at enterprise scale.
- Led containerization of legacy monolithic Java applications using Docker, resolving environment configuration drift across dev, staging, and release pipelines, and reducing production deployment bugs by 15% improving release stability significantly across systems platform.

PROJECTS

Sorex — Android Offline AI Assistant

- Built fully offline Android AI assistant using Phi 3.5 mini with MLC LLM and GGUF quantization, integrating AES 256 encryption and vector indexing to enable retrieval, document Q&A, and sub second inference without cloud dependency reduced latency.

Clinical Retrieval Acceleration System

- Improved clinical document retrieval by replacing keyword search with RAG pipeline using pgvector semantic search and FastAPI, reducing latency from 2.3s to 1.5s while eliminating hallucinations via grounding evaluation metrics sub second endpoints production ready

EDUCATION

University of New Haven

Master of Science (M.S.), Data Science Aug 2023 - Present

Sreenidhi Institute of Science and Technology

Bachelor of Technology (B.Tech) Jun 2017 - May 2021

CERTIFICATIONS

- Oracle Cloud Infrastructure Multicloud Architect
- Oracle Analytics Cloud (OAC)
- OCI AI & Data Platform